1. bu yerda $[a] - a\ sonining\ butun\ qismi$

$$[-\sqrt{20}] + [-\sqrt{21}] + [-\sqrt{22}] + \cdots + [-\sqrt{40}]$$

A) −124 B) −105 C) −62 D) −50

2. $\frac{2+\sqrt{3}}{\sqrt{2}+\sqrt{2+\sqrt{3}}} + \frac{2-\sqrt{3}}{\sqrt{2}-\sqrt{2-\sqrt{3}}} =$

A) 2 B) $\sqrt{2}$ C) $\frac{\sqrt{2}}{2}$ D) $\frac{\sqrt{6}}{3}$

3. $4+2\sqrt{2}$ somiga teskari sonni toping.

A) $4-2\sqrt{2}$ B) $-4-2\sqrt{2}$ C) $0{,}5-0{,}25\sqrt{2}$ D) $\frac{1}{4-2\sqrt{2}}$

4. Hisoblang:

$$\frac{\sqrt{7}+\sqrt{5}}{\sqrt{28}+\sqrt{21}+\sqrt{20}+\sqrt{15}} - \frac{\sqrt{3}+\sqrt{5}}{\sqrt{20}-\sqrt{15}+\sqrt{12}-3}$$

A) 2 B) −2 C) $2\sqrt{3}$ D) $-2\sqrt{3}$

5. ikki sonning yig`indisi $\sqrt{6}\ ga$, ayirmasi $\sqrt{10}$ ga teng. ularning ko`paytmasi ikkidan qanchaga kam?

A) 1 B) 2 C) 3 D) 4

6. x=$\sqrt{2}$ va y=$\sqrt{5}$ bo`lsa,

$$\sqrt{9x^2+12xy+4y^2} - \sqrt{9x^2-12xy+4y^2} = ?$$

A) $-6\sqrt{2}$ $B)\ 4\sqrt{5}$ $C) -4\sqrt{5}$ $D)\ 6\sqrt{2}$

7. Hisoblang: $\frac{\sqrt{5-3\sqrt{2,(6)}}}{\sqrt{2}} + \frac{\sqrt{5+3\sqrt{2,(6)}}}{\sqrt{2}}$

A) 2 B) $\sqrt{6}$ C) $2\sqrt{3}$ D) $2\sqrt{2}$

8. x= n+$\sqrt{n^2-16}$; $y = n - \sqrt{n^2-16}$ va y=2 bo`lsa, x va y ning o`rta arifmetigini toping.

A) 3 B) 4 C) 5 D) 8

9. Hisoblang: $\sqrt{\frac{13-6\sqrt{4,(6)}}{3+6\sqrt{0,(22)}}} - \sqrt{\frac{13+6\sqrt{4,(6)}}{3-6\sqrt{0,(22)}}}$

A) $2\sqrt{7}$ B) $2(2\sqrt{3}-\sqrt{7})$ C) $-2(\sqrt{7}+2\sqrt{3})$ D) $2\sqrt{14}$

10.Soddalashtiring:

$$\frac{x\sqrt{y}-3y\sqrt{x}-4y\sqrt{y}}{x\sqrt{x}+4x\sqrt{y}+3y\sqrt{x}} : \frac{y\sqrt{y}-x\sqrt{y}}{x\sqrt{x}+2x\sqrt{y}-3y\sqrt{x}}$$

A) $\frac{3\sqrt{y}+\sqrt{x}}{\sqrt{y}-\sqrt{x}}$ B) $\frac{\sqrt{y}-3\sqrt{x}}{\sqrt{y}+\sqrt{x}}$ C) $\frac{\sqrt{y}+\sqrt{x}}{\sqrt{y}-\sqrt{x}}$ D) $\frac{4\sqrt{y}-\sqrt{x}}{\sqrt{y}+\sqrt{x}}$

11. $a = \pi - e$, bo`lsa ifodani soddalashtiring.

$$\sqrt{a^2(a^2-a+1)+a^2(a+1)+1} - \sqrt{a^2(a^2+a-1)-a^2(a+1)+1}$$

A) $2-2a^2$ $B)\ 2a^2$ $C)\ 2$ $D)\ \sqrt{a^4+1}-\sqrt{a^4-1}$

12. Hisodblang:

$$\sqrt{11}\cdot\left(\frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \cdots + \frac{1}{\sqrt{98}+\sqrt{99}} + 1\right)$$

A) 11 B) 22 C) 33 D) 44

13. $\sqrt[3]{4\sqrt{2\sqrt[3]{4\sqrt{2\ldots}}}} + \sqrt{30+\sqrt{30+\sqrt{30+\cdots}}} = ?$

A) 6 B) 7 C) 8 D) 9

14. $\frac{\sqrt{10+\sqrt{1}}+\sqrt{10+\sqrt{2}}+\cdots+\sqrt{10+\sqrt{99}}}{\sqrt{10-\sqrt{1}}+\sqrt{10-\sqrt{2}}+\cdots+\sqrt{10-\sqrt{99}}}$ ni hisoblang.

A) 1 B) 2 C) $\sqrt{2}-1$ D) $\sqrt{2}+1$

15. $\sqrt{8+2\sqrt{10+2\sqrt{5}}} + \sqrt{8-2\sqrt{10+2\sqrt{5}}} = ?$

A) $\sqrt{2}+\sqrt{5}$ B) $\sqrt{2}+\sqrt{3}$ C) $\sqrt{2}+\sqrt{10}$ D) $\sqrt{2}+1$

16. $\frac{\sqrt{0,5}}{\sqrt{2,4}}\cdot\left(\sqrt{\frac{1,2-0,7}{1,2+0,7}} + \sqrt{\frac{2,4+1,4}{2,4-1,4}}\right)\cdot\frac{\sqrt{1,5+0,4}}{\sqrt{0,9+1,5}} = ?$

A) 1 B) 2 C) $\sqrt{24}$ D) 5

17. $\sqrt{\frac{n^4+2n^3+2n^2+2n+1}{n^2(n+1)^2}}$ ni soddalashtiring

A) $1+\frac{1}{n}-\frac{1}{n+1}$ B) $1+\frac{1}{n}+\frac{1}{n+1}$

C) $1-\frac{1}{n}-\frac{1}{n+1}$ D) $1-\frac{1}{n}+\frac{1}{n+1}$

18. Hisoblang: $\sqrt[3]{5+2\sqrt{13}} + \sqrt[3]{5-2\sqrt{13}}$

A) $\sqrt[3]{2}$ B) $0{,}25\sqrt[3]{65}$ C) 1 D) 1,5

19. $x^2+y^2+z^2 = 6x+8y+10z-50$ bo`lsa, x ni toping.

A) 27 B) cheksiz ko`p C) 1 D) 3

20. Agar $\sqrt{4\sqrt{2}+2\sqrt{6}} = \sqrt[4]{a}+\sqrt[4]{b}$ bo`lsa, a+b ni toping.

A) 20 B) 18 C) 16 D) 10

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21. Bunda x=8 va y= $2\sqrt{2}$

$$\frac{x+y}{\sqrt[3]{x^2}-\sqrt[3]{xy}+\sqrt[3]{y^2}}\cdot(x^{\frac{1}{3}}-y^{\frac{1}{3}})\cdot$$

$$\frac{x-y}{\sqrt[3]{x^2}+\sqrt[3]{xy}+\sqrt[3]{y^2}}\cdot(x^{\frac{1}{3}}+y^{\frac{1}{3}})=?$$

A) 4 B) 5 C) 6 D) 8

22. Hisoblang:

$$\frac{3}{2\sqrt[3]{2}+2\sqrt[3]{4}}-\frac{3}{2\sqrt[3]{2}-2\sqrt[3]{4}}+\sqrt[3]{4}$$

A) $\sqrt[3]{4}$ B) $\sqrt[3]{16}+1$

C) $(\sqrt[3]{4}+1)^2$ D) $-(\sqrt[3]{4}+1)^2$

23. $\sqrt{22-30\sqrt{4-2\sqrt{3}}}+5+3\sqrt{3}$

A) 10 B) $6\sqrt{3}$ C) 12 D) $5-3\sqrt{3}$

24. $\frac{\sqrt{1}\cdot\sqrt{2}\cdot\sqrt{3}\cdot\ldots\cdot\sqrt{10}}{\sqrt{x}}$ kasr ratsional son bo`lishi uchun x quyidagilarning qaysi biriga teng.

A) 7 B) 5 C) 3 D) 2

25. $\sqrt{\sqrt{241+44\sqrt{30}}}-(\sqrt{6}-\sqrt{5})^{-1}$

A) 0 B) 1 C) $\sqrt{30}$ D) 11

26. Agar a=39$-\sqrt{432}$ bo`lsa, $\sqrt{a}+\sqrt{3}$ ifodaning qiymatini toping.

A) 6 B) 3 C) $6+\sqrt{3}$ D) $6\sqrt{3}$

27. Hisoblang:

$$\left(\frac{10}{\sqrt{6}+1}+\frac{2}{\sqrt{6}-2}-\frac{6}{3-\sqrt{6}}\right)\cdot\left(\sqrt{96}+\sqrt{\left(\frac{1}{24}\right)^{-2}}\right)$$

A) −480 B) −240 C) −120 D) −60

28. Soddalashtiring:

$$\frac{(a+2\sqrt{a}+1)(\sqrt{a}+\sqrt{b})(\sqrt{a}-\sqrt{b})}{(a-b)(\sqrt{a}+1)^2}+2$$

A) 0 B) 1 C) -1 D) 3

29. Hisoblang:

$$\left(5^{5\sqrt{5}}\cdot\sqrt{5^{5-10\sqrt{5}}}\right)^{\left(\frac{\sqrt{5}}{2}\right)^{-2}}$$

A) 1 B) $\sqrt{4}$ C) 5 D) 25

30. A= $\sqrt{2}-\sqrt[3]{2}$, $B=\sqrt[3]{2}+\sqrt[6]{2}+1$ va $C=\sqrt{2}-1$ bo`lsa, $\frac{AB}{C}$ ning qiymatini toping.

A) $\sqrt[6]{2}$ B) $\sqrt[3]{2}$ C) $\sqrt{2}$ D) 2

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O`quvchilarim mazza qilsin: 2-Variant

1. x= n+$\sqrt{n^2-16}$; $y=n-\sqrt{n^2-16}$ va y=2 bo`lsa, x va y ning o`rta arifmetigini toping.

A) 3 B) 4 C) 5 D) 8

2. Hisoblang: $\sqrt{\frac{13-6\sqrt{4,(6)}}{3+6\sqrt{0,(22)}}}-\sqrt{\frac{13+6\sqrt{4,(6)}}{3-6\sqrt{0,(22)}}}$

A) $2\sqrt{7}$ B) $2(2\sqrt{3}-\sqrt{7})$ C) $-2(\sqrt{7}+2\sqrt{3})$ D) $2\sqrt{14}$

3.Soddalashtiring:

$$\frac{x\sqrt{y}-3y\sqrt{x}-4y\sqrt{y}}{x\sqrt{x}+4x\sqrt{y}+3y\sqrt{x}}:\frac{y\sqrt{y}-x\sqrt{y}}{x\sqrt{x}+2x\sqrt{y}-3y\sqrt{x}}$$

A) $\frac{3\sqrt{y}+\sqrt{x}}{\sqrt{y}-\sqrt{x}}$ B) $\frac{\sqrt{y}-3\sqrt{x}}{\sqrt{y}+\sqrt{x}}$ C) $\frac{\sqrt{y}+\sqrt{x}}{\sqrt{y}-\sqrt{x}}$ D) $\frac{4\sqrt{y}-\sqrt{x}}{\sqrt{y}+\sqrt{x}}$

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A) $-6\sqrt{2}$ $B)\ 4\sqrt{5}$ $C)-4\sqrt{5}$ $D)\ 6\sqrt{2}$

5. Hisoblang: $\frac{\sqrt{5-3\sqrt{2,(6)}}}{\sqrt{2}}+\frac{\sqrt{5+3\sqrt{2,(6)}}}{\sqrt{2}}$

A) 2 B) $\sqrt{6}$ C) $2\sqrt{3}$ D) $2\sqrt{2}$

6. $\frac{\sqrt{0,5}}{\sqrt{2,4}}\cdot\left(\sqrt{\frac{1,2-0,7}{1,2+0,7}}+\sqrt{\frac{2,4+1,4}{2,4-1,4}}\right)\cdot\frac{\sqrt{1,5+0,4}}{\sqrt{0,9+1,5}}$ =?

A) 1 B) 2 C) $\sqrt{24}$ D) 5

7. $\sqrt{\frac{n^4+2n^3+2n^2+2n+1}{n^2(n+1)^2}}$ ni soddalashtiring

A) $1+\frac{1}{n}-\frac{1}{n+1}$ B) $1+\frac{1}{n}+\frac{1}{n+1}$

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8. Hisoblang: $\sqrt[3]{5+2\sqrt{13}}+\sqrt[3]{5-2\sqrt{13}}$

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16. Agar a=39−$\sqrt{432}$ bo`lsa, $\sqrt{a}+\sqrt{3}$ ifodaning qiymatini toping.

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18. Soddalashtiring:

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19. A= $\sqrt{2}-\sqrt[3]{2}$, $B=\sqrt[3]{2}+\sqrt[6]{2}+1$ va $C=\sqrt{2}-1$ bo`lsa, $\frac{AB}{C}$ ning qiymatini toping.

A) $\sqrt[6]{2}$ B) $\sqrt[3]{2}$ C) $\sqrt{2}$ D) 2

20. Hisoblang:

$$(5^{5\sqrt{5}}\cdot\sqrt{5^{5-10\sqrt{5}}})^{(\frac{\sqrt{5}}{2})^{-2}}$$

A) 1 B) $\sqrt{4}$ C) 5 D) 25

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$$-\sqrt{a^2(a^2+a-1)-a^2(a+1)+1}$$

A) $2-2a^2$ B) $2a^2$ C) 2 D) $\sqrt{a^4+1}-\sqrt{a^4-1}$

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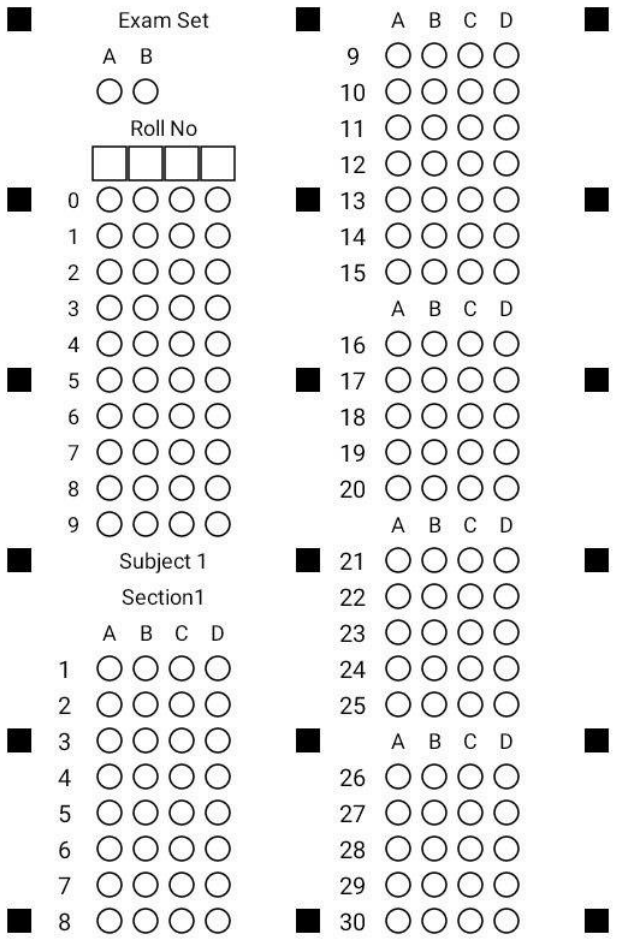
A) $\sqrt{2}+\sqrt{5}$ B) $\sqrt{2}+\sqrt{3}$ C) $\sqrt{2}+\sqrt{10}$ D) $\sqrt{2}+1$



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